UNIT 6 · RAPID REVISION

Formula and Fact Sheet

Soil strength and failure

Item	Value
Mohr–Coulomb shear strength	$\tau = c + \sigma \cdot \tan \varphi$ (c = intercept, φ = slope)
Mohr failure condition	Mohr circle tangent to the failure envelope
Plate/lug maximum force	$F = A \cdot c + W \cdot \tan \varphi$
Tyre footprint area	$A \approx 0.78 \cdot b \cdot l$ (track: $A = b \cdot l$)
Sand	$c \approx 0$ — thrust ∝ load; ballast helps
Saturated clay	$\varphi \approx 0$ — $\tau = c$; ballast only sinks
Plasticity index	PI = LL - PL
Cone index standard	ASAE S313.2 — 30° apex cone
Cone index averaging depth	0–15 cm (for sinkage < 7.5 cm); measure before traffic
Brittle vs plastic soil	Dense/dry: peak then residual; loose/wet: work-hardening flow

Tillage tools

Item	Value
Practical rake angle (narrow tools)	~25–45°
Narrow tine below critical depth	Lateral compaction, no useful loosening
Dynamic draft term	$\propto v^2$ (Goryachkin)
Reece equation terms	Soil weight, cohesion, surcharge, adhesion
Nichols — plastic-soil shear	Peaks at the lower plastic limit; zero at the upper (liquid) limit
Soil–steel μ vs clay content	~Independent above C ≈ 32%
Polished steel	Cuts μ by 5–20% in high-clay soils, not sand
Down / side suction	⅛–⅜ in — holds depth / width

Traction and compaction

Item	Value
Net pull	$H = F - MR$; $\mu = \mu_g - \rho$
Slip	$S = 1 - V_a/V_t$ ($V_t = r\omega$; ASAE S296.2)
Working slip	8–15% firm soil; ~20% tilled
Bekker sinkage	$p = (k_c/b + k_\varphi)\cdot z^n$
Wheel numeric	$Cn = C\cdot b\cdot d/W$
Wismer–Luth gross traction	$\mu_g = 0.75(1 - e^{(-0.3\cdot Cn\cdot S)})$
Wismer–Luth motion resistance	$\rho = 1.2/Cn + 0.04$
Design example (Cn = 30)	TE 0.72, H/W ≈ 0.51 (chart), slip 0.16
Tyre 13.6–38	Section width 13.6 in, rim diameter 38 in
First-pass compaction	71–91% (bulk density), 74–96% (sinkage) of the 4-pass total
Root-limiting bulk density	~1.6–1.8 g/cm³ (cone index > ~2 MPa)
Axle load vs contact pressure	Axle load → depth; contact pressure → surface severity

Soil properties and failure (Topics 1–2)

- Soil dynamics (Gill & Vanden Berg, USDA Hbk 316, 1967) = soil behaviour under tool/traction forces; DYNAMIC (being worked) vs static (survey) properties.
- Mohr–Coulomb: $\tau = c + \sigma\cdot\tan\varphi$; c = intercept (cohesion), φ = slope (angle of internal friction); failure when the Mohr circle is tangent to the envelope.
- Plate/lug (Liljedahl): $F = Ac + W\cdot\tan\varphi$; plot F vs W → intercept Ac, slope tanφ; tyre footprint $A \approx 0.78\, b\cdot l$; track $A = b\cdot l$.
- Sand: c = 0 (weight-dependent thrust); saturated clay: φ ≈ 0 (weight-independent, τ = c). Adhesion and soil–metal friction δ govern tool scouring.
- Cone index (ASAE S313.2, 30° cone) = force/base area to push a cone at a steady rate; 0–15 cm average; measure before traffic; brittle (dense/dry) soil peaks then softens, plastic (wet) soil work-hardens and smears.

Tillage tools (Topics 3–4)

- Tillage tool = moving wedge/inclined plane → compresses soil → shear failure (Kepner p. 143); the shear reaction spreads either side of the plane (cohesion + interlocking).
- Rake angle α is the master variable; draft rises steeply beyond the optimum (~25–45° for tines); narrow tine → forward crescent shear; below the CRITICAL DEPTH → lateral compaction.
- Nichols: plastic-soil shear peaks near the lower plastic limit, zero at the upper; μ (soil–steel) ~independent of clay above C ≈ 32%; polishing cuts μ 5–20% in high-clay soils, not sand.
- Adhesion → poor scouring (wet black soils); cure with polished/coated surfaces. Wear blunts the edge and loses suction → higher draft; use hard-facing and disposable shares.
- Down suction (vertical) + side suction (lateral) ~⅛–⅜ in hold depth/width (Kepner p. 142). Performance: specific draft, specific energy, MWD, inversion, scouring, wear — soil bins + field.

Traction (Topics 5–8)

- Traction = driving force of a wheel/track; TE = output/input power; μ = H/W; ρ = MR/W; μ = μ_g − ρ; H = F − MR.
- Slip S = 1 − V_a/V_t (V_t = rω); a wheel must slip to pull; working slip 8–15% firm, ~20% tilled; ASAE S296.2 rolling radius.
- Cn = C·b·d/W; Wismer–Luth H/W = 0.75(1 − e^(−0.3·Cn·S)) − (1.2/Cn + 0.04); μ_g = 0.75(1 − e^(−0.3·Cn·S)).
- TE peaks at low slip (lower slip for higher Cn); design slip right of the TE peak for pull; Cn = 30 → TE 0.72, H/W 0.51, slip 0.16.
- Bekker sinkage p = $(k_c/b + k_\varphi)z^n$. Tyre 13.6–38 = width–rim (in); R-1 general / R-2 cane-rice / R-3 turf / R-4 industrial; radial > bias for TE; lower inflation and bigger footprint improve traction and cut compaction.

Soil compaction (Topic 9)

- Compaction = densification (bulk density ↑, porosity ↓); failure by COMPRESSION (vs shear under a tine); largely irreversible without tillage.
- Axle load → DEPTH of compaction; contact pressure (inflation/tyre width) → SEVERITY at the surface; wet soil most compactible; FIRST pass does 71–96% of the damage (Liljedahl Table 10-3, p. 257).
- Proctor: maximum bulk density at an OPTIMUM moisture — soil most vulnerable near field capacity; keep heavy traffic off wet soil.
- Effects: macroporosity, infiltration, aeration ↓; strength, penetration resistance, tillage draft ↑; plough pan forms; roots restricted above ~1.6–1.8 g/cm³ (cone index > ~2 MPa). Seed-zone firming (press wheel) is good.
- Alleviation: reduce ground pressure (radial/dual/low inflation/tracks), controlled traffic, fewer passes, avoid wet traffic, organic matter; break pans by subsoiling when the subsoil is DRY. Prevention > cure.

Further reading

- Gill, W. R. & Vanden Berg, G. E. — Soil Dynamics in Tillage and Traction, USDA Agriculture Handbook No. 316 (1967).
- Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10 (Traction).
- Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 6–7.
- Wismer, R. D. & Luth, H. J. (1974) — Off-road traction prediction for wheeled vehicles, Trans. ASAE.
- ASAE S313.2 (cone penetrometer), S296.2 (traction terminology), D497 (machinery management data).